

# Julian C. Stamm

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## EDUCATION

- Oct 2025 – present    **University of Bonn**, M.Sc. in Computer Science
- Focus on Real-time GI, Neural Rendering, and Machine Learning
- Oct 2021 – Sept 2025    **University of Bonn**, B.Sc. in Computer Science
- Grade: 1.1 ([top 3%](#) [↗](#))
  - Bachelor's Thesis: Bidirectional Neural Radiance Caching [\[full-res\]](#) [↗](#)

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## PUBLICATIONS

- May 2026    **Photon-Driven Neural Radiance Caching**
- Continuation of Bachelor's Thesis "Bidirectional Neural Radiance Caching" with focus on Photon trained NRC. Presented at I3D 2026.
- Julian C. Stamm, Tom Kneiphof, Reinhard Klein  
[10.1145/3807895.3807923](https://doi.org/10.1145/3807895.3807923) [↗](#) (I3D Companion '26)

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## EXPERIENCE

- Feb 2026 – Mar 2026    **Teaching Assistant for "Foundations of 4D/6D Object Capture for Virtual Environments"**, University of Bonn – Bonn, Germany
- Held daily block tutorials on Nerfs, IK, 3DGS, differentiable rendering etc.
- Apr 2024 – present    **Teaching Assistant for "Introduction to Computer Graphics"**, University of Bonn – Bonn, Germany
- Held weekly tutorials on rasterization, ray tracing, BSDFs, splines etc.
  - Assisted in creation of exercises and exams
  - Developed and maintained modern OpenGL/C++ framework used in the exercises and the animation competition
- May 2021 – Sept 2021    **Temporary Worker in Software Development Team**, Atlas Spare Parts GmbH – Ganderkesee, Germany
- Developed custom CAN bus logger.

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## PROJECTS

- May 2026 – present    [tsnn](#) [↗](#) (**unpublished**)
- High performance Slang-native tiny-cuda-nn alternative with Slangs auto-differentiation.
- Implementation of neural spline flows, hash grid encodings, MLPs, Adam, and more
  - Achieves full kernel fusion with Falcor path tracer
  - Slightly outperforms tiny-cuda-nn in my NRC benchmark
  - Written for my current lab research on real-time neural importance sampling
- Feb 2025 – present    [PhotonNRC](#) [↗](#)
- Falcor implementation of my Bachelor Thesis "Bidirectional Neural Radiance Caching" and poster "Photon-Driven Neural Radiance Caching".
- Hardware-accelerated SPPM implementation
  - NRC implementation using tiny-cuda-nn
  - Implementation of Photon-NRC

Feb 2025 – Sept 2025 [BiNRC](#) ↗

OptiX implementation of my Bachelor Thesis "Bidirectional Neural Radiance Caching".

- Hardware-accelerated SPPM implementation
- NRC implementation using tiny-cuda-nn
- Unbiased path tracer with MIS and NEE
- Implementation of basic Disney principled BSDF
- Implementation of Photon-NRC

Aug 2024 – present [nBounce](#) ↗

Self-educational real-time path tracer with software BVH construction and traversal.

- Written in Rust and wgpu

Oct 2023 – Mar 2024 [Stable Real-Time Simulation of Mesoscopic Glints](#) ↗

Computer Graphics Lab as part of the B.Sc. Computer Science.

- Implementation of the discrete stochastic microfacet model from "Real-Time Rendering of Glinty Appearances Using Distributed Binomial Laws on Anisotropic Grids" by Thomas Deliot and Laurent Belcour

Aug 2023 – present [gltemplate](#) ↗

OpenGL 4.6 framework developed for CG exercises.

- Written in C++ and CMake, RAII-based resource management, texture/mesh loading utilities

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## OTHERS

Apr 2026 – present [bevy\\_trail](#) ↗

Bevy plugin for high performance GPU-based trails in wgpu.

Sept 2025 – present **P2P Multiplayer Game**

Working on a Python-moddable P2P-multiplayer card game in Rust using bevy.

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## SKILLS

**Languages:** C++, CMake, GLSL, CUDA, Slang, Python, Rust, WGSL, Java

**APIs:** OpenGL, OptiX, Falcor, wgpu, bevy, Godot, PyTorch

**Research Interests:** Real-Time GI, Neural Rendering, Path Tracing, Photon Mapping, Machine Learning, Normalizing Flows

**Spoken Languages:** German (native), English (TOEFL iBT: 110/120)